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**Lai**

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(54) **MEMORY DEVICE WITH UNIPOLAR SELECTOR**

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See application file for complete search history.

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Hsin-Chu (TW)

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This patent is subject to a terminal disclaimer.

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**Related U.S. Application Data**

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(57)

**ABSTRACT**

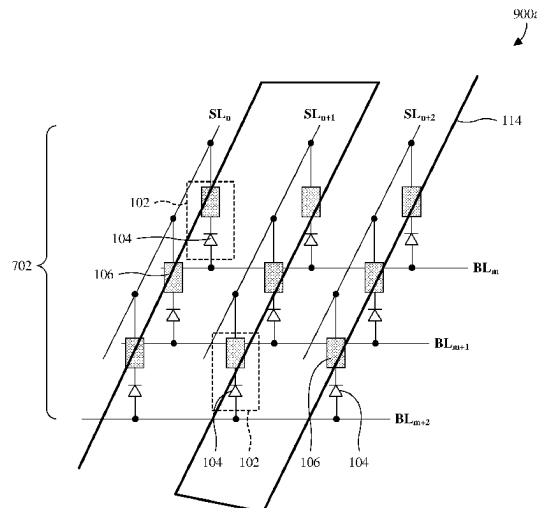
(51) **Int. Cl.**  
**G11C 11/16** (2006.01)  
**G11C 5/06** (2006.01)

Various embodiments of the present application are directed towards a method of forming a memory device. The method includes forming a lower part of an interconnect structure over a substrate and forming a unipolar selector over the lower part of the interconnect structure. The method further comprises forming a data-storage element over the unipolar selector and electrically coupled in series with the unipolar selector, the data-storage element having a variable resistance. The method further comprises generating an external magnetic field by a magnetic field generator to pre-set the data-storage element to a first data state.

(52) **U.S. Cl.**  
CPC ..... **G11C 11/161** (2013.01); **G11C 5/06** (2013.01); **G11C 11/1659** (2013.01); **G11C 11/1673** (2013.01); **G11C 11/1675** (2013.01); **G11C 11/1697** (2013.01)

(58) **Field of Classification Search**  
CPC ..... G11C 11/161; G11C 11/1659; G11C 11/1673; G11C 11/1675; G11C 11/1697; G11C 5/06

**20 Claims, 15 Drawing Sheets**



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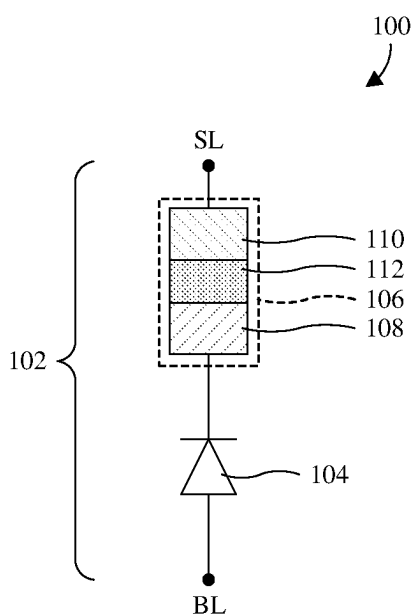
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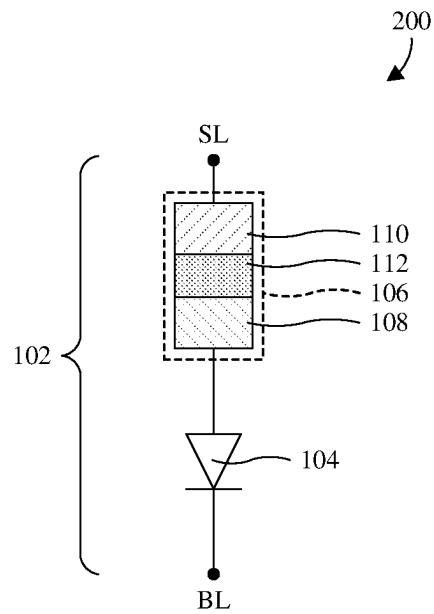
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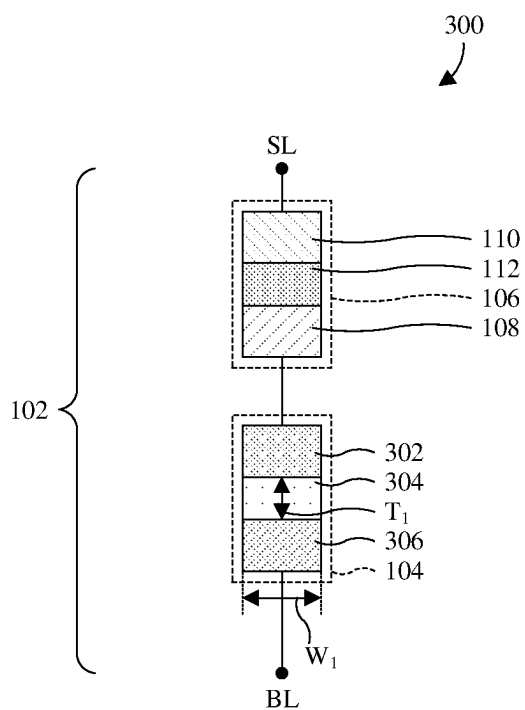
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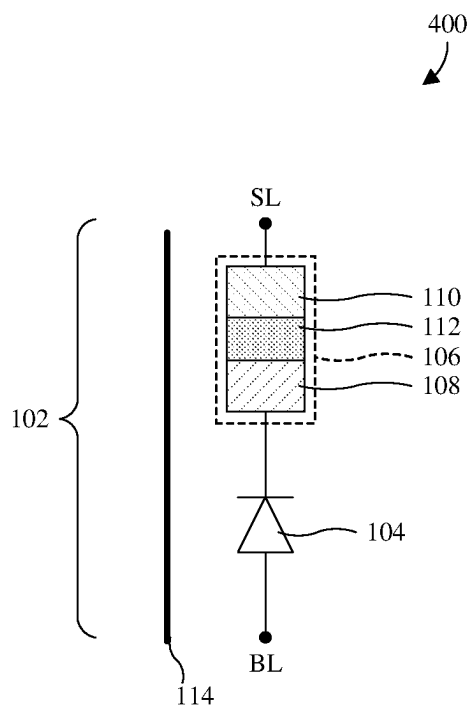
**Fig. 1**



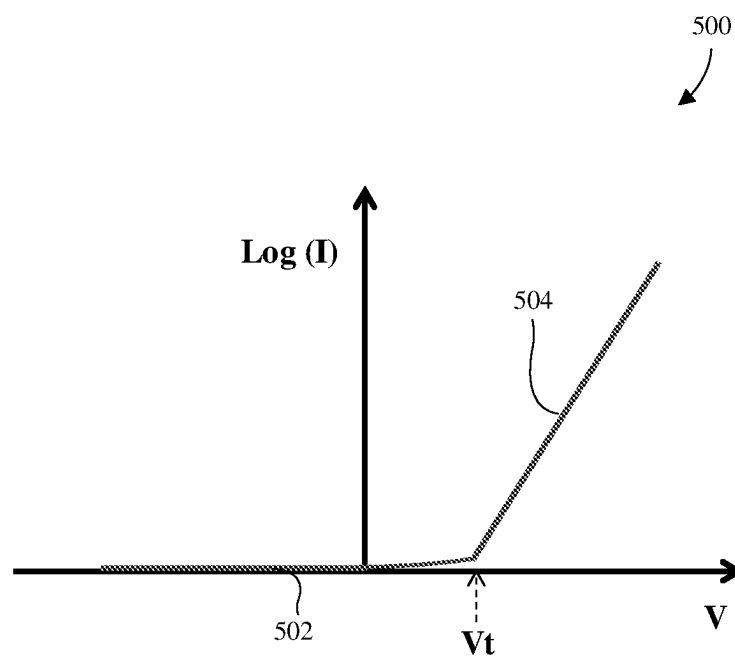
**Fig. 2**



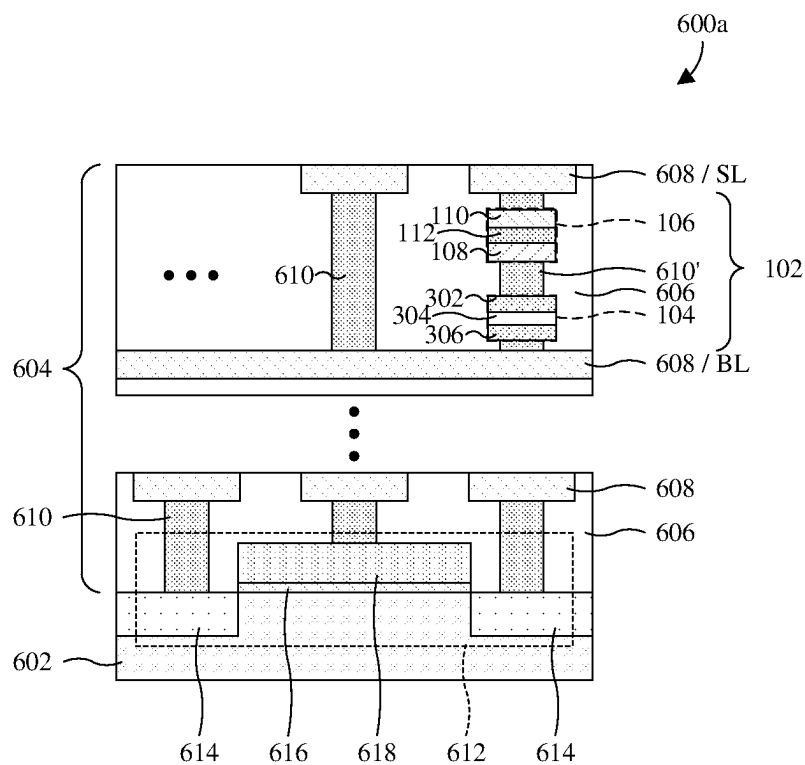
**Fig. 3**



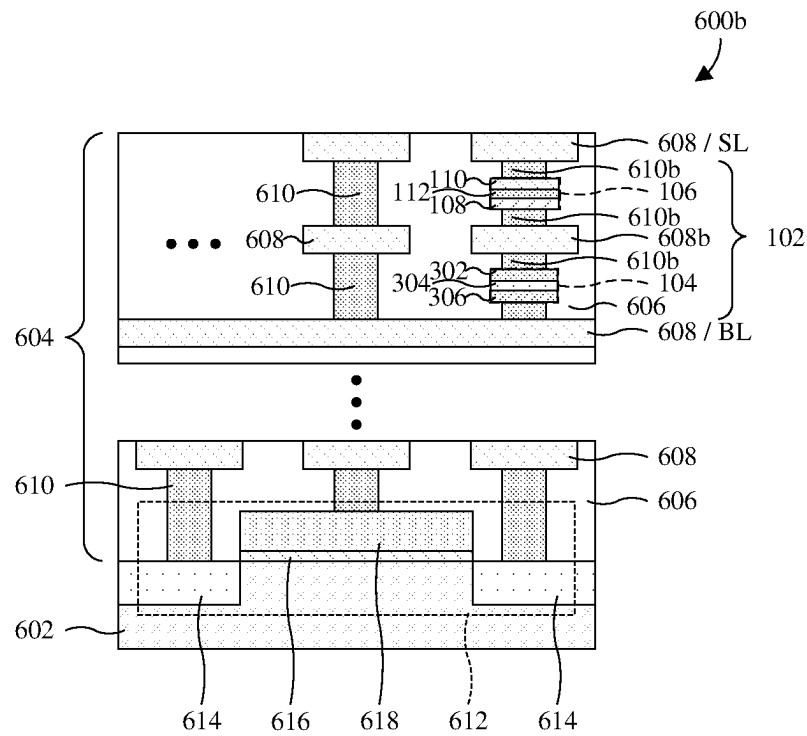
**Fig. 4**



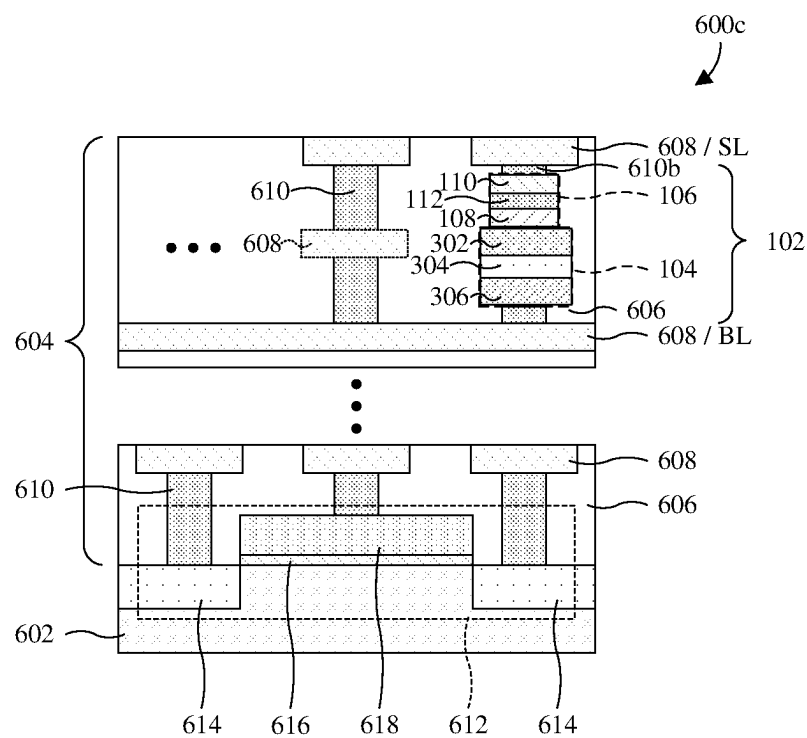
**Fig. 5**



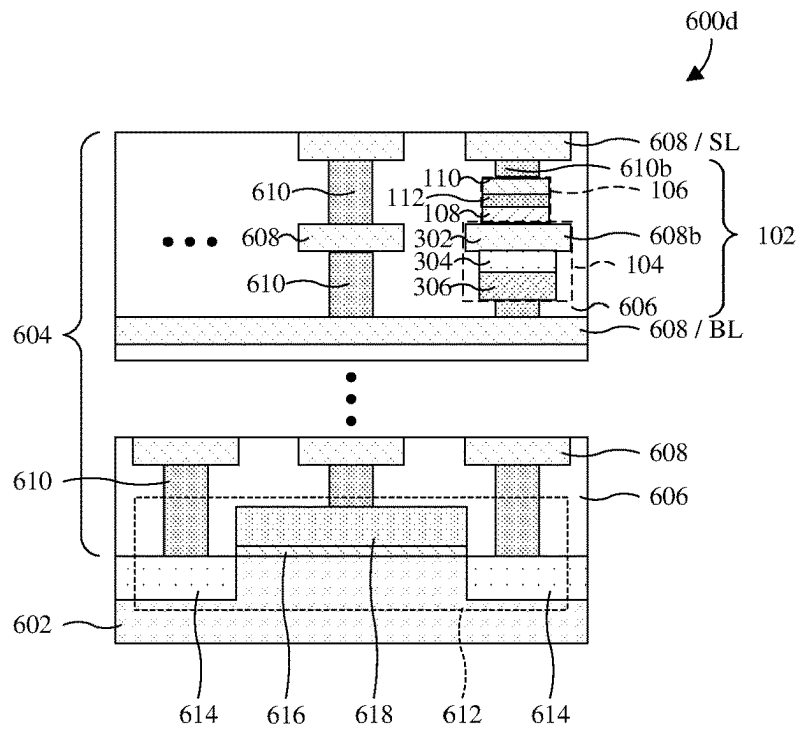
**Fig. 6A**



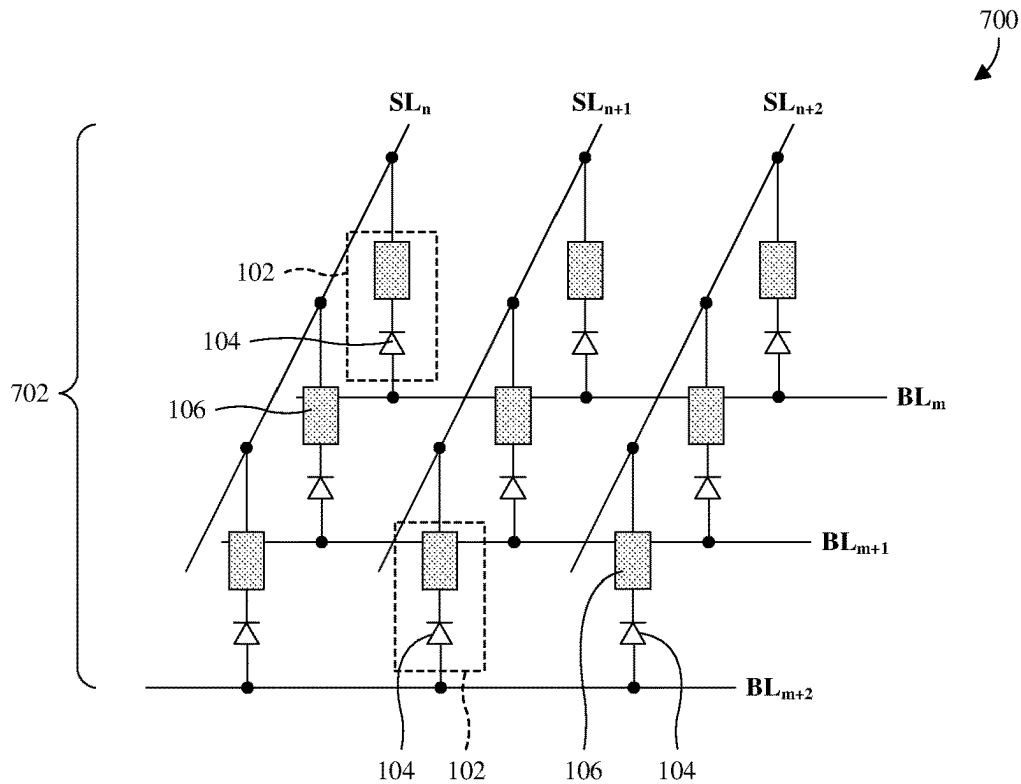
**Fig. 6B**



**Fig. 6C**



**Fig. 6D**



**Fig. 7**

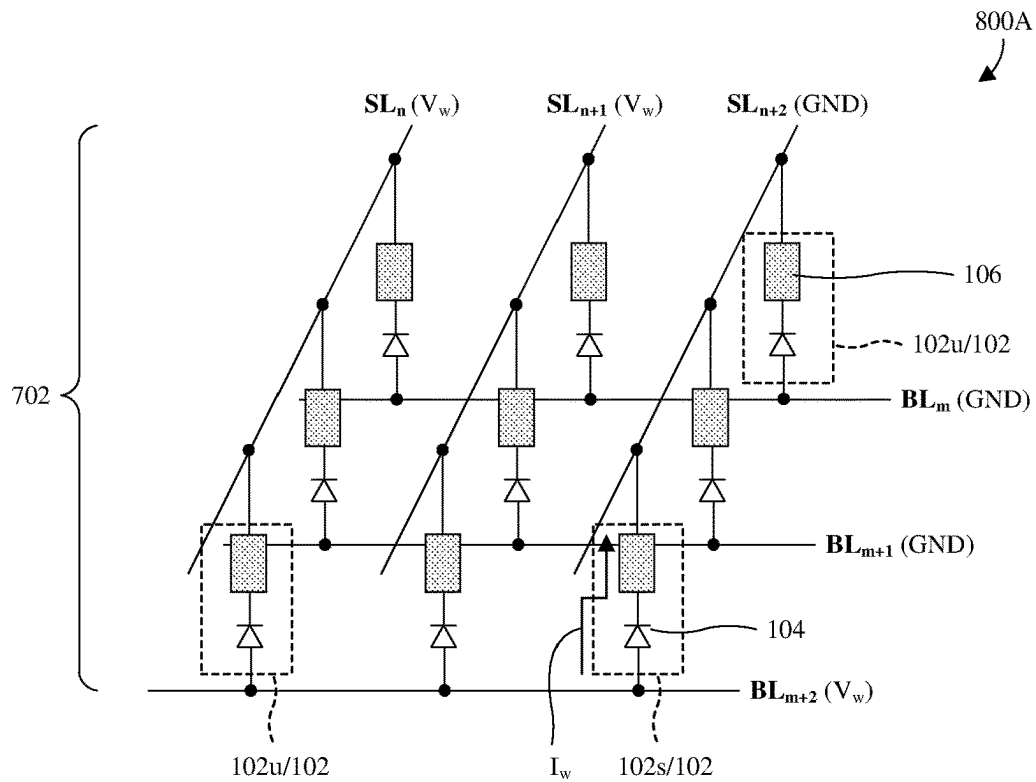


Fig. 8A

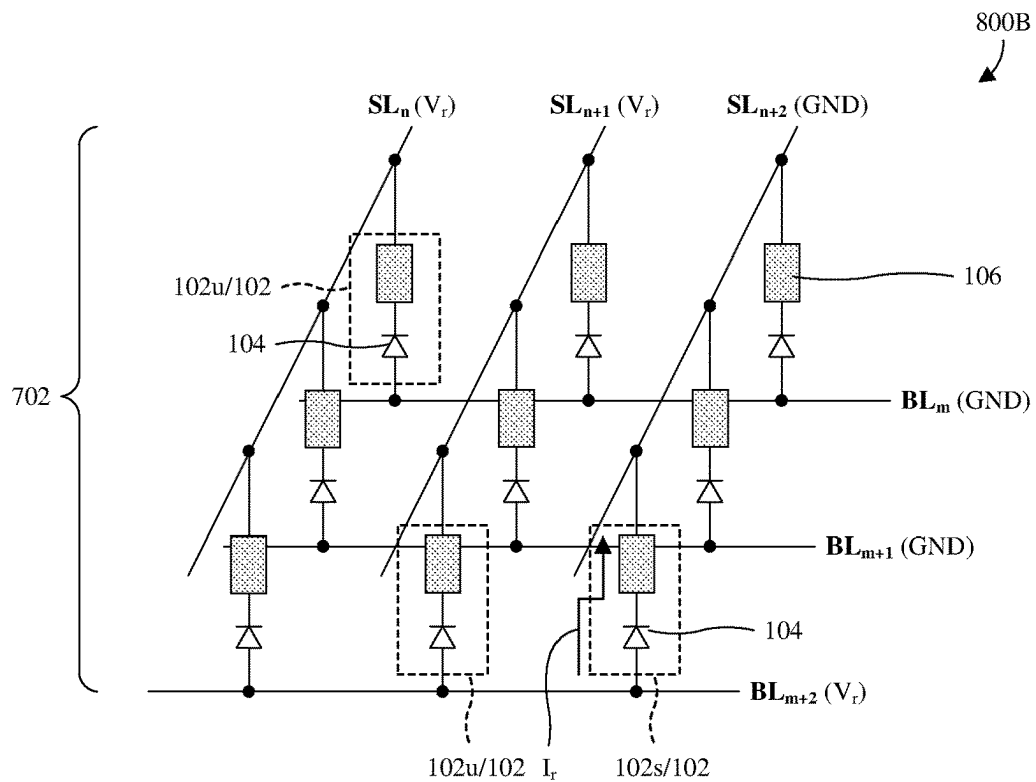


Fig. 8B

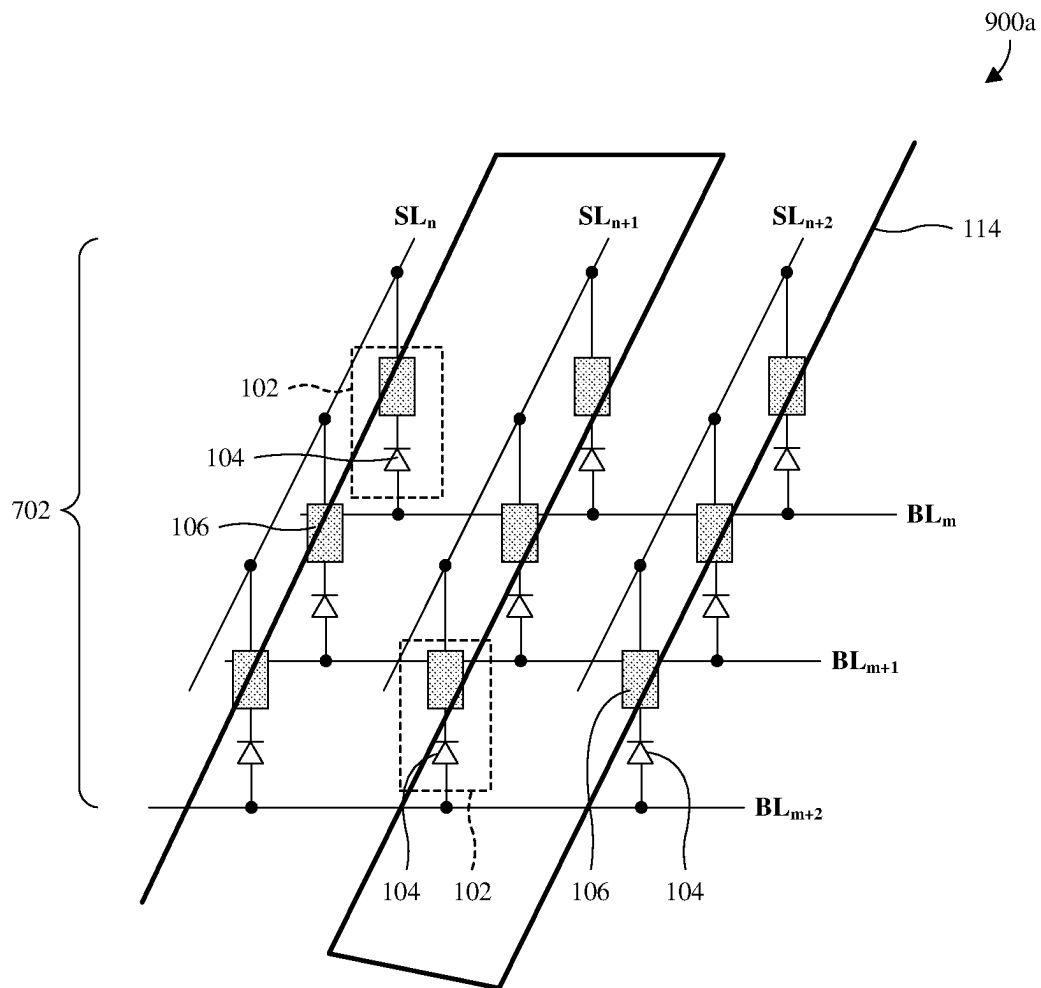
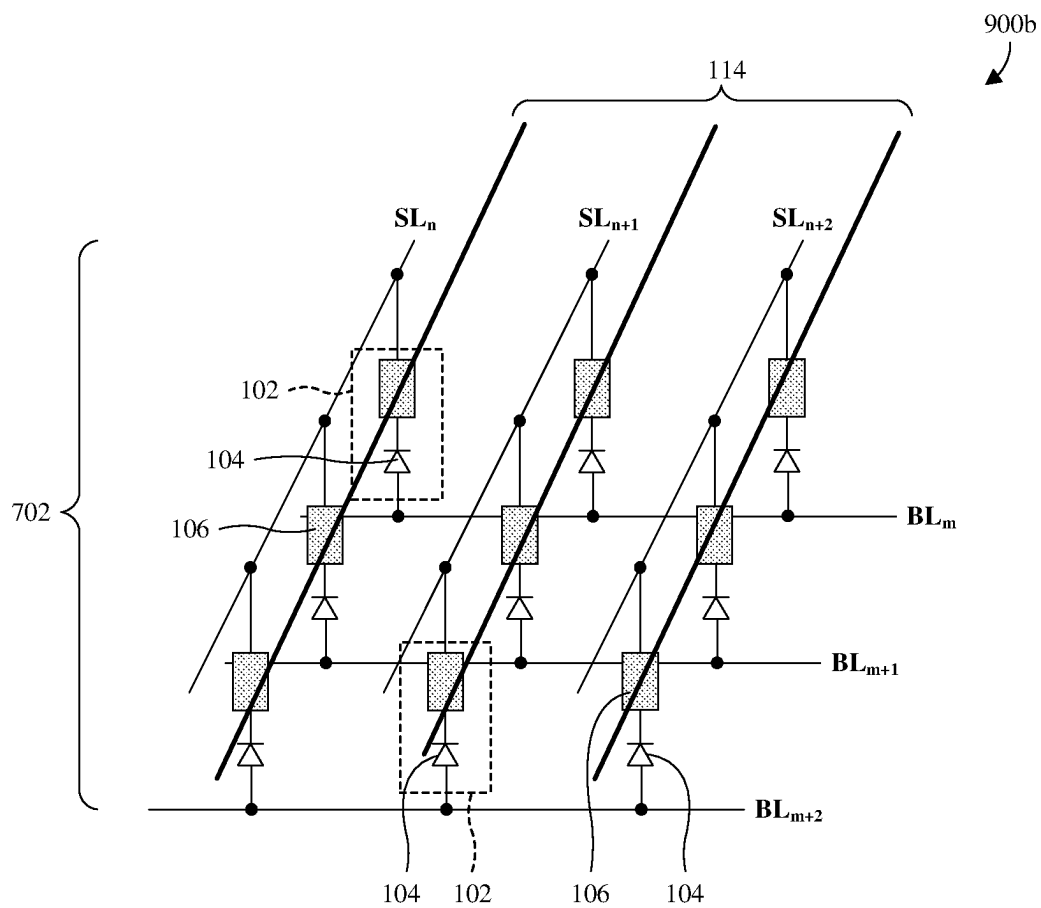


Fig. 9A





**Fig. 9B**

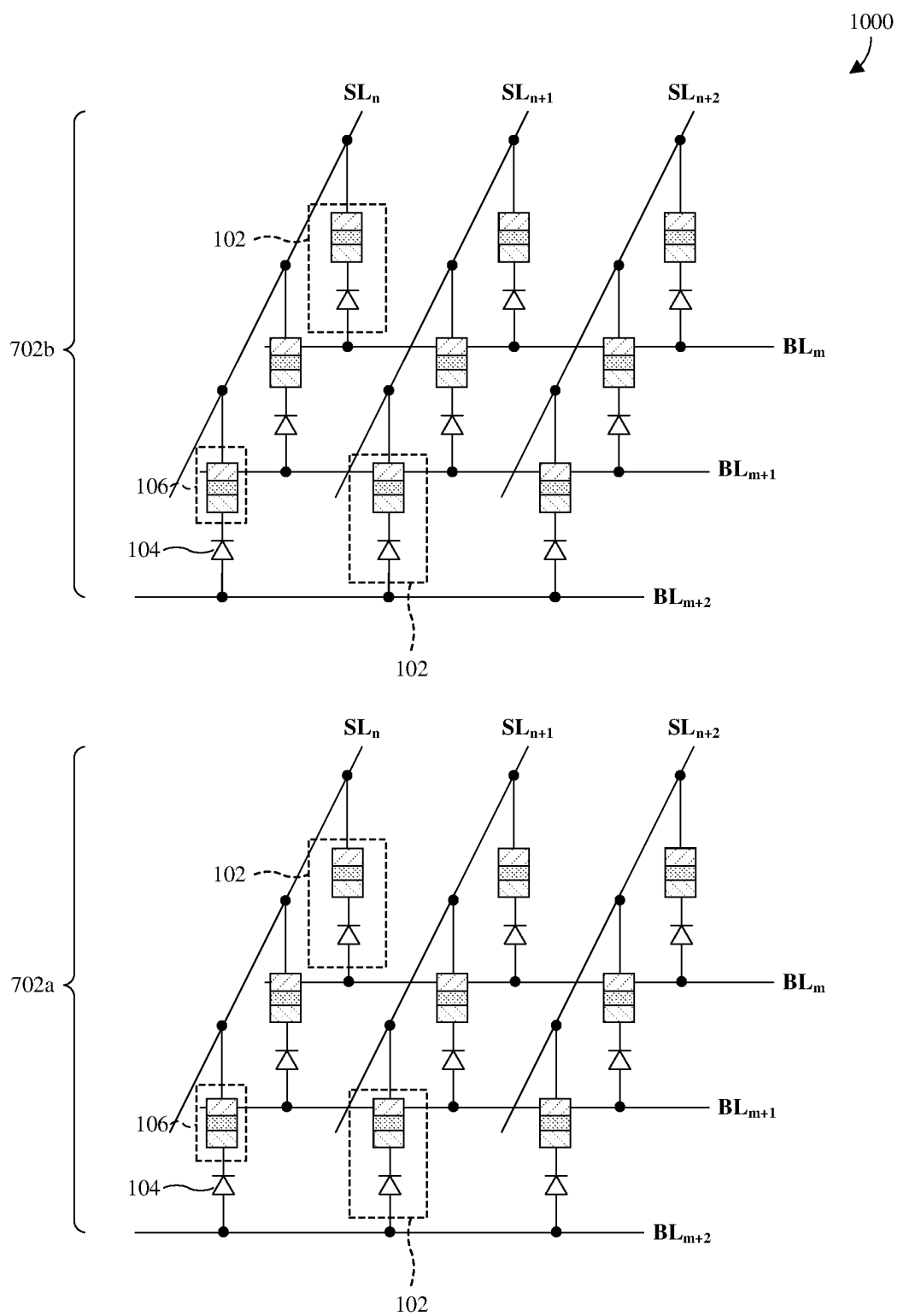


Fig. 10

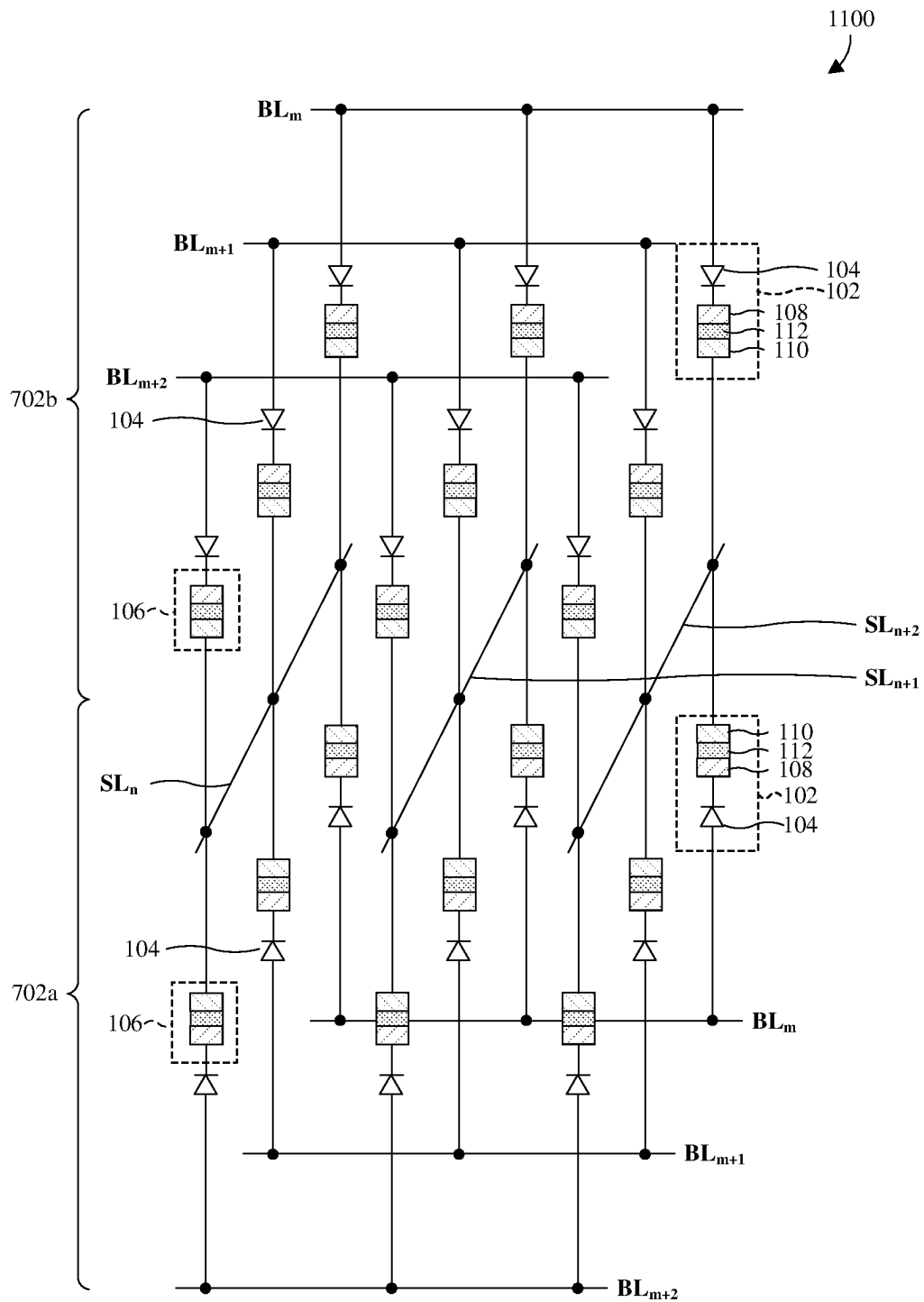


Fig. 11

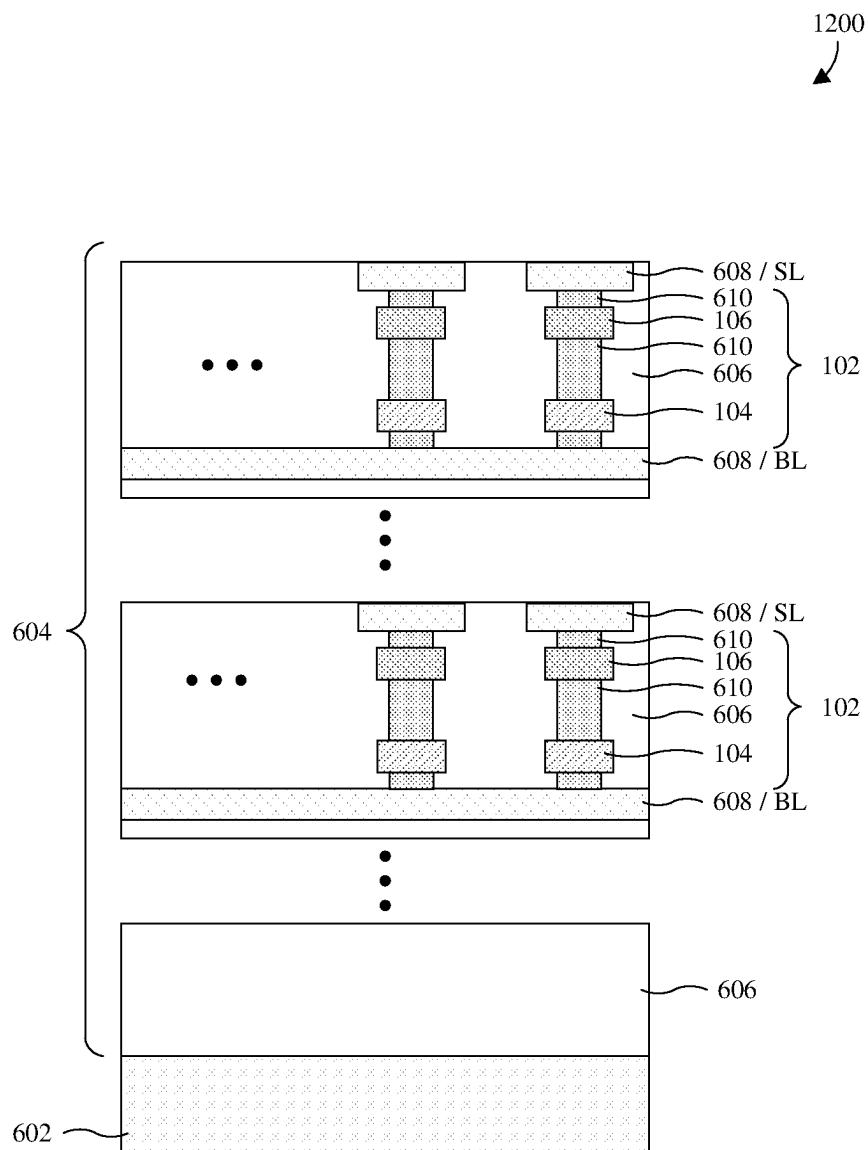
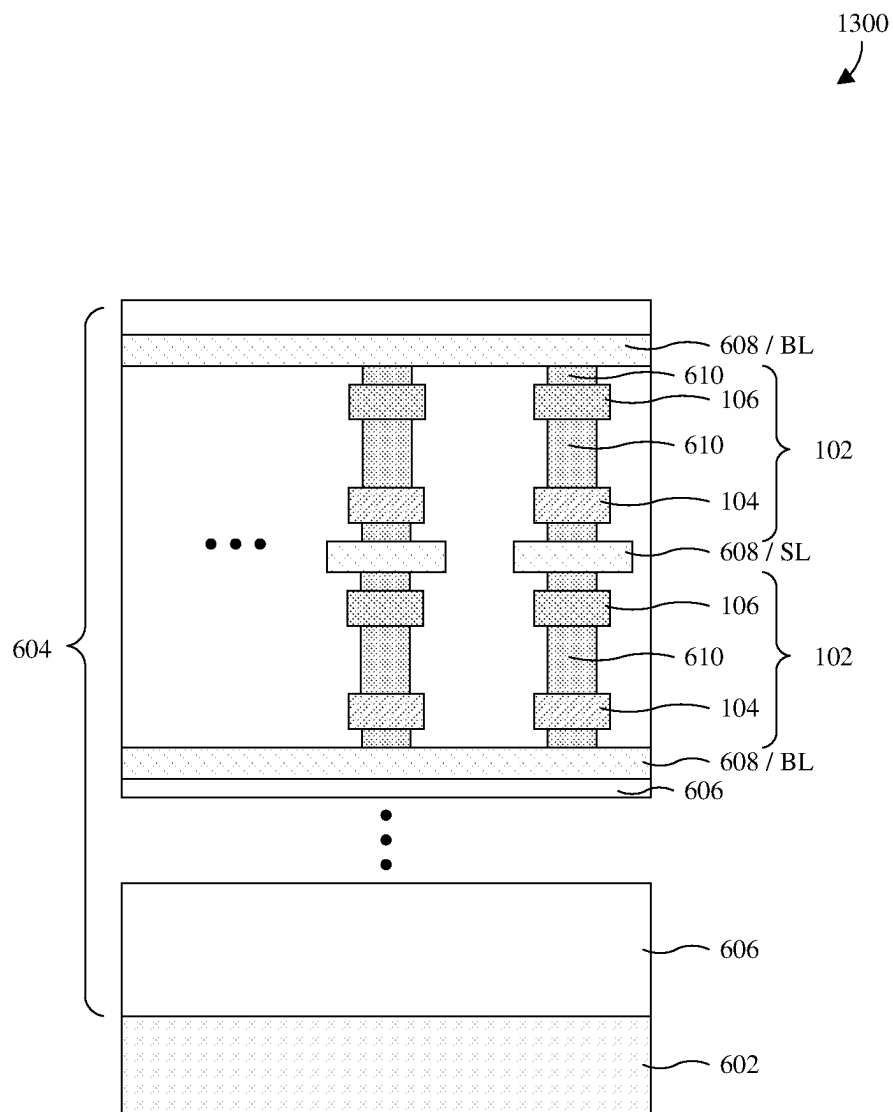
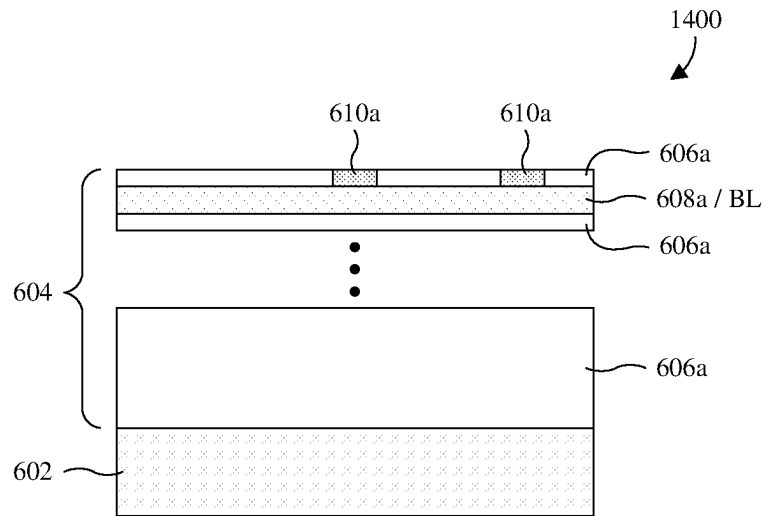


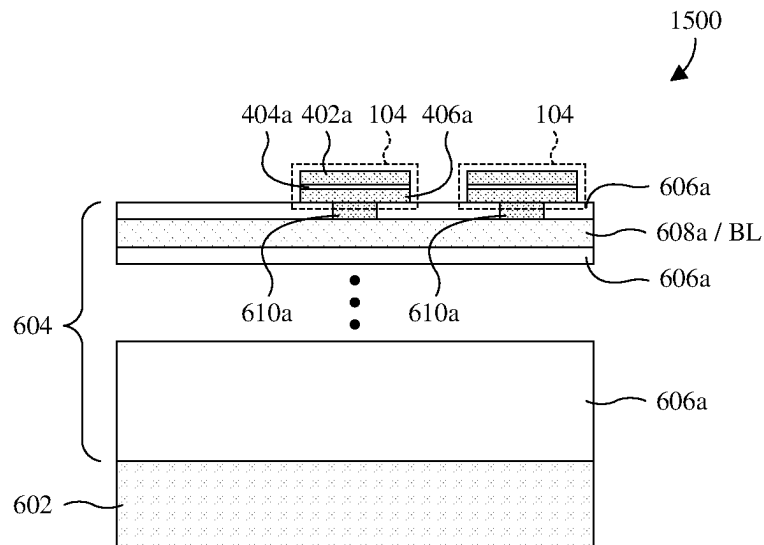
Fig. 12



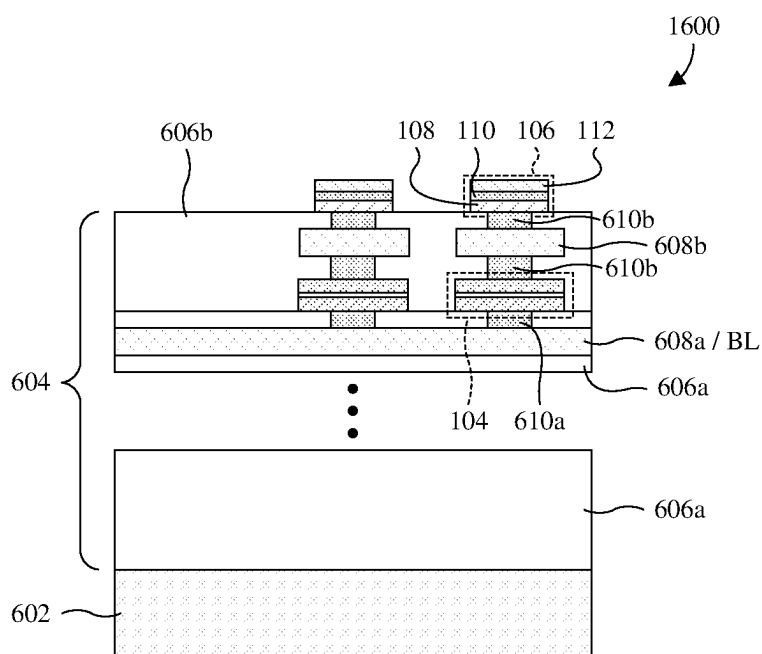
**Fig. 13**



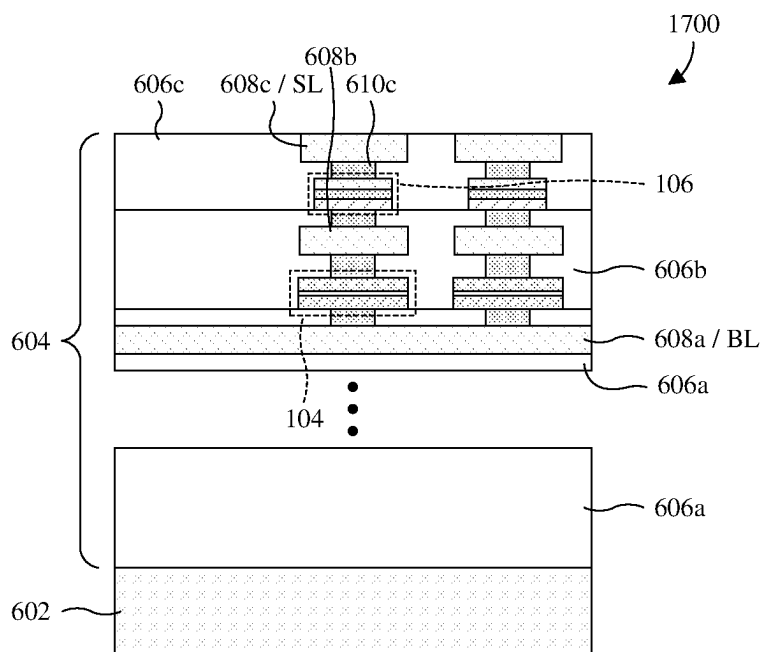
**Fig. 14**



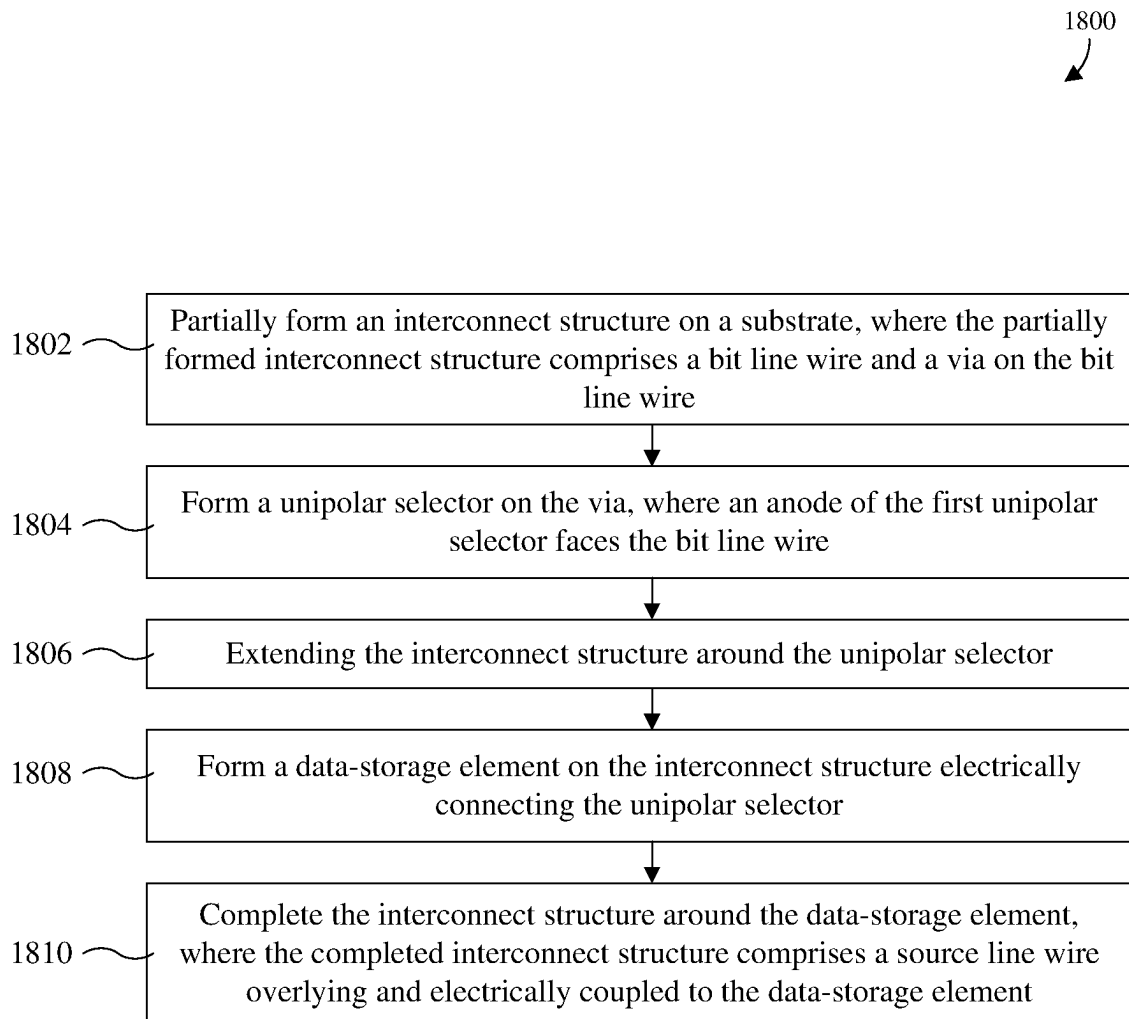
**Fig. 15**



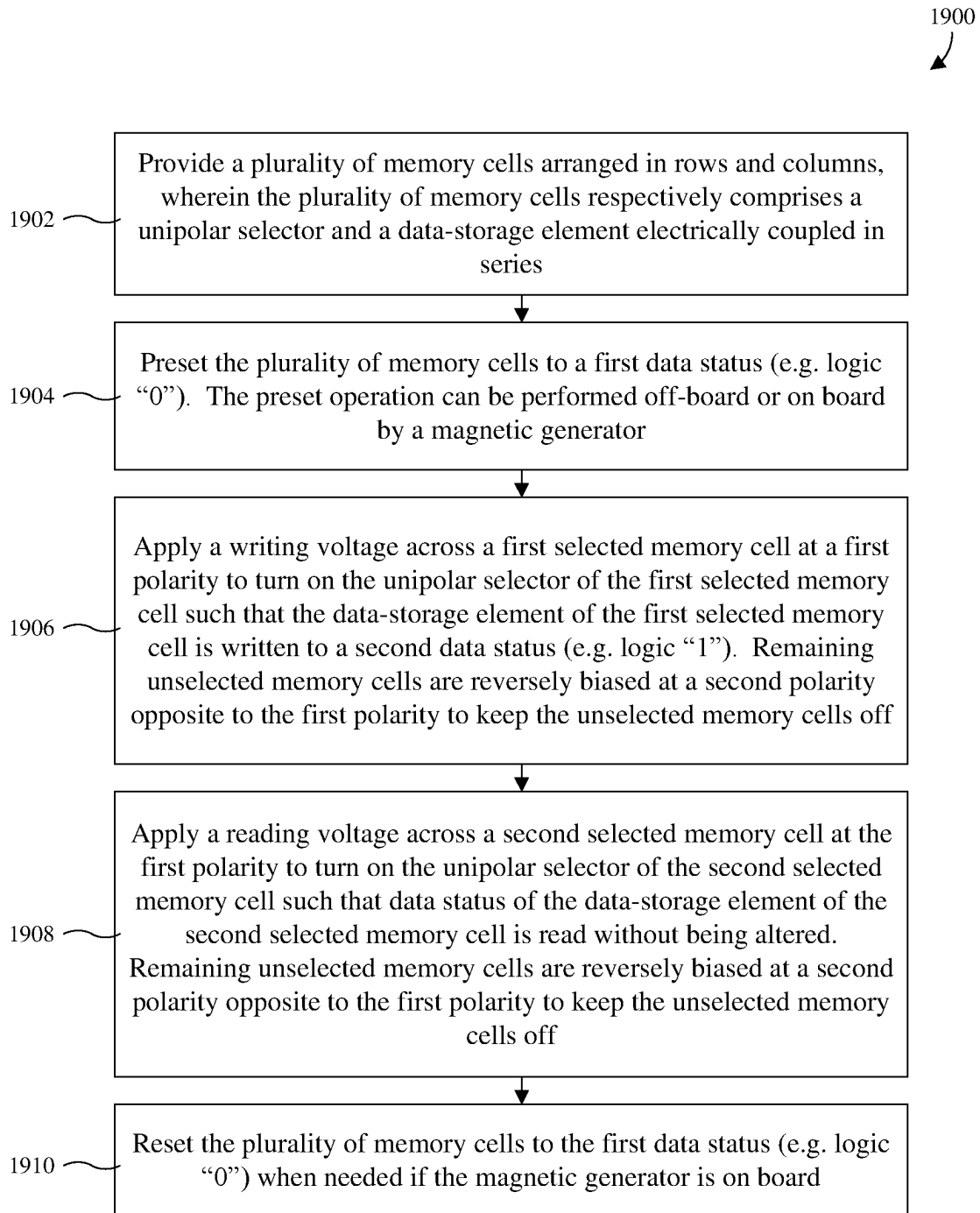
**Fig. 16**



**Fig. 17**

**Fig. 18**



**Fig. 19**

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## MEMORY DEVICE WITH UNIPOLAR SELECTOR

### REFERENCE TO RELATED APPLICATION

This application is a Divisional of U.S. application Ser. No. 16/908,914, filed on Jun. 23, 2020, the contents of which are hereby incorporated by reference in their entirety.

### BACKGROUND

Many modern-day electronic devices include electronic memory. Electronic memory may be volatile memory or non-volatile memory (NVM). Non-volatile memory is able to store data in the absence of power, whereas volatile memory is not. Non-volatile memory such as magnetoresistive random-access memory (MRAM) and resistive random access memory (RRAM) are promising candidates for next generation non-volatile memory technology due to relative simple structures and their compatibility with complementary metal-oxide-semiconductor (CMOS) logic fabrication processes.

### BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a schematic diagram of some embodiments of a memory cell comprising a unipolar selector.

FIG. 2 illustrates a schematic diagram of some alternative embodiments of a memory cell comprising a unipolar selector.

FIG. 3 illustrates a schematic diagram of some more detailed embodiments of the memory cell of FIG. 1 in which the unipolar selector comprises a multilayer stack.

FIG. 4 illustrates a schematic diagram of some embodiments of the memory cell of FIG. 1 comprising a unipolar selector and a magnetic field generator coupled to the memory cell.

FIG. 5 illustrates a graph of some embodiments of current-voltage (I-V) curves for the unipolar selector of FIG. 1.

FIGS. 6A-6D illustrate cross-sectional views of various embodiments of an integrated chip comprising the memory cell of FIG. 1, FIG. 2, FIG. 3, or FIG. 4.

FIG. 7 illustrates a schematic view of some embodiments of a memory array comprising a plurality of memory cells of FIG. 1, FIG. 2, or FIG. 3.

FIGS. 8A-8B illustrate schematic views of some embodiments of the memory array of FIG. 7 at various operational states.

FIGS. 9A-9B illustrate schematic views of various embodiments of the memory array comprising a plurality of memory cells of FIG. 4.

FIGS. 10 and 11 illustrate schematic views of various embodiments of a three dimensional (3D) memory array comprising stacked memory arrays of memory cells of FIG. 1, FIG. 2, FIG. 3 or FIG. 4.

FIGS. 12 and 13 illustrate cross-sectional views of various embodiments of an integrated chip comprising the three dimensional (3D) memory array in FIGS. 10 and 11.

FIGS. 14-17 illustrate a series of cross-sectional views of some embodiments of a method for forming an integrated

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chip comprising a memory array, where memory cells of the memory array comprise unipolar selectors.

FIG. 18 illustrates a block diagram of some embodiments of the method of FIGS. 14-17.

FIG. 19 illustrates a block diagram of some embodiments of a method of operating a memory device that can include a memory cell of FIGS. 1-4, a memory array of FIGS. 7-11, or an integrated chip to FIG. 6A-6D, 12 or 13.

### DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

A cross-point memory architecture with one-selector one-resistor (1S1R) memory cells is increasingly receiving attention for use with next generation electronic memory due to its high density. A cross-point memory array may, for example, comprise multiple one-selector one-resistor (1S1R) memory cells respectively arranged at cross points of bit lines and source lines. The selector is a bipolar device configured to pass bidirectional current when biased above respective threshold voltages. By appropriately biasing a bit line and a source line (e.g. BL0 and SL0), a 1S1R memory cell at a cross point of the bit line and the source line can be selected and written to opposite states. When a 1S1R memory cell is selected, other bit lines and source lines may be biased at a middle point voltage to turn off unselected memory cells. However, a first group of unselected memory cells shares the same bit line (BL0) with the selected 1S1R memory cell and thus is biased at a difference voltage of the bit line voltage and the middle point voltage. Similarly, a second group of unselected memory cells share the same source line (SL0) with the selected 1S1R memory cell and thus is biased at a difference voltage of the source line voltage and the middle point voltage. The collective leakage current flowing through the first group of unselected memory cells and the second group of unselected memory cells introduces disturbance and reduces the current window for memory operation for reading and writing operations.

The disturbance may even result in a reading failure during the reading operation or a false writing during the writing operation.

In view of above, various embodiments of the present application are directed towards a memory cell using a unipolar selector, as well as a unipolar operation method of such memory cell. The unipolar selector may, for example, be a diode or some other suitable unipolar device that is turned on when a forward bias greater than its threshold voltage is applied. The unipolar selector is electrically coupled in series with a data-storage element and controls current flowing through the data-storage element or voltage applied across the data-storage element. In some embodiments, the memory cell is read from and written to at a single polarity and cannot be rewritten. In some alternative embodiments, a reset operation may be performed by other means such as using an external magnetic field generated by an off-board or on-board magnetic field generator. By using the unipolar selector rather than the bipolar selector, unselected memory cells can be biased at the opposite polarity and thus minimize leakage current and reduce reading and writing disturbance.

With reference to FIG. 1, a schematic diagram 100 of some embodiments of a memory cell 102 comprising a unipolar selector 104 is provided. A unipolar selector switches at a single polarity whereas a bipolar selector switches at two polarities. At a first polarity, the unipolar selector conducts and/or is in a low resistance state called "on" state if the voltage across the unipolar selector exceeds a threshold voltage. Otherwise, at the first polarity, the unipolar selector is non-conducting or is in a high resistance state called "off" state. At the second polarity, the unipolar selector is in the "off" state. The unipolar selector 104 is configured to selectively allow current to flow in a first direction from a bit line BL to a source line SL, while blocking the flow of current in a second direction from the source line SL to the bit line BL. In some embodiments, the unipolar selector 104 has only two terminals. In some alternative embodiments, the unipolar selector 104 has more than two terminals. The unipolar selector 104 may, for example, be PIN diodes, polysilicon diodes, punch-through diodes, varistor-type selectors, ovonic threshold switches (OTSs), doped-chalcogenide-based selectors, Mott effect based selectors, mixed-ionic-electronic-conductive (MIEC)-based selectors, field-assisted-superlattice-threshold (FAST) selectors, filament-based selectors, doped-hafnium-oxide-based selectors, or some other suitable diodes and/or selectors.

The unipolar selector 104 is electrically coupled in series with a data-storage element 106, from the bit line BL to the source line SL. In some embodiments, locations of the bit line BL and the source line SL can be reversed. An example of the operation is as follows: when the voltage across the unipolar selector 104 is positive from the bit line BL to the data-storage element 106, the unipolar selector 104 conducts and is in a low resistance state if the voltage across the unipolar selector 104, from the bit line BL to the data-storage element 106, exceeds a threshold voltage  $V_t$ . Otherwise, the unipolar selector 104 is non-conducting and/or is in a high resistance state. The data-storage element 106 stores a bit of data. As an example, during the writing operation, a writing voltage is applied such that the unipolar selector 104 is biased above the threshold voltage at the first polarity and the data-storage element 106 is set to a first data state. During the reading operation, a reading voltage is applied such that the unipolar selector 104 is biased above the threshold voltage at the first polarity while the data-

storage element 106 is not altered. The reading voltage may be smaller than the writing voltage.

In some embodiments, a resistance of the data-storage element varies depending upon a data state of the data-storage element 106. For example, the data-storage element 106 may have a low resistance at a first data state and may have a high resistance at a second data state. In other embodiments, capacitance or some other suitable parameter of the data-storage element 106 varies depending upon a data state of the data-storage element 106. In some embodiments, the data-storage element 106 is a magnetic tunnel junction (MTJ) or a spin-valve and is written by a spin-transfer torque (STT) method. In such cases, the memory cell 102 is referred as a STT magnetic memory cell, and the memory device made of an array of such memory cells is referred as a STT-MRAM device. The STT method is described in more details below. In some alternative embodiments, the data-storage element 106 is a metal-insulator-metal (MIM) stack, and the memory cell 102 may be a resistance memory cell. Other structures for the data-storage element 106 and/or other memory-cell types for the memory cell 102 are also amenable.

As an example, the data-storage element 106 comprises a reference ferromagnetic element 108, a free ferromagnetic element 110, and a barrier element 112 and stores data using the STT method. The barrier element 112 is non-magnetic and is sandwiched between the reference ferromagnetic element 108 and free ferromagnetic element 110. The reference ferromagnetic element 108 has a fixed magnetization, whereas the free ferromagnetic element 110 has variable a magnetization. The barrier element 112 can be a tunneling barrier layer. In some embodiments, a spin-polarized current is created by passing a current through the reference ferromagnetic element 108. This current is then directed into free ferromagnetic elements 110, transfers the angular momentum, and changes the spin of the electrons in the free ferromagnetic elements 110. Magnetic moments of the reference ferromagnetic element 108 and the free ferromagnetic element 110 can be in-plane or perpendicular to a silicon substrate surface. Devices with perpendicular magnetic moments are more scalable compared to those with in-plane moments and are also more cost competitive. Depending upon whether magnetizations of the reference ferromagnetic element 108 and free ferromagnetic element 110 are parallel or anti-parallel, the data-storage element 106 has a low resistance or a high resistance. For example, the data-storage element 106 may have the low resistance when the magnetizations of the reference ferromagnetic element 108 and free ferromagnetic element 110 are parallel and may have the high resistance when the magnetizations are anti-parallel. The low and high resistances may, in turn, be used to represent different data states of the data-storage element 106.

In some embodiments, the barrier element 112 is a tunnel barrier selectively allowing quantum mechanical tunneling of electrons through the barrier element 112. For example, quantum mechanical tunneling may be allowed when the reference ferromagnetic element 108 and free ferromagnetic element 110 have parallel magnetizations, and may be blocked when the reference ferromagnetic element 108 and free ferromagnetic element 110 have antiparallel magnetizations. The barrier element 112 may, for example, be or comprise an amorphous barrier, a crystalline barrier, or some other suitable barrier. The amorphous barrier may be or comprise, for example, aluminum oxide (e.g.,  $\text{AlO}_x$ ), titanium oxide (e.g.,  $\text{TiO}_x$ ), or some other suitable amorphous barrier. The crystalline barrier may be or comprise manga-

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nese oxide (e.g., MgO), spinel (e.g.,  $\text{MgAl}_2\text{O}_4$ ), or some other suitable crystalline barrier.

In some embodiments, the reference ferromagnetic element **108** is or comprises cobalt iron (e.g., CoFe), cobalt iron boron (e.g., CoFeB), or some other suitable ferromagnetic material(s), or any combination of the foregoing. In some embodiments, the reference ferromagnetic element **108** adjoins an antiferromagnetic element (not shown) and/or is part of or otherwise adjoins a synthetic antiferromagnetic (SAF) element (not shown). In some embodiments, the free ferromagnetic element **110** is or comprises cobalt iron (e.g., CoFe), cobalt iron boron (e.g., CoFeB), or some other suitable ferromagnetic material(s), or any combination of the foregoing.

With reference to FIG. 2, a schematic diagram **200** of some alternative embodiments of the memory cell **102** of FIG. 1 is provided in which the reference ferromagnetic element **108** overlies the free ferromagnetic element **110**. Since the reference ferromagnetic element **108** overlies the free ferromagnetic element **110**, the polarities at which the writing voltage is applied across the data-storage element **106** are reversed compared to FIG. 1. The writing voltage is applied across the data-storage element **106** at the second polarity to set the data-storage element **106** to the antiparallel state.

With reference to FIG. 3, a schematic diagram **300** of some more detailed embodiments of the memory cell **102** of FIG. 1 is provided in which the unipolar selector **104** is a multilayer stack. The unipolar selector **104** comprises a cathode **302**, an insulator **304**, and an anode **306**. The insulator **304** is sandwiched between the cathode **302** and the anode **306**. In some embodiments, the cathode **302** is directly connected to the reference ferromagnetic element **108** of the data-storage element **106**, meaning the cathode **302** is electrically connected to the reference ferromagnetic element **108** by one or more conductive wires and/or vias without other electronic devices disposed therebetween. In some alternative embodiments, the unipolar selector **104** may be reversely placed that the anode **306** is directly connected to the reference ferromagnetic element **108**. The multilayer stacks may, for example, be or comprise a PIN diode or some other multilayer devices. In some embodiments in which the multilayer stack is a PIN diode, the cathode **302** is or comprises N-type semiconductor material, the anode **306** is or comprises P-type semiconductor material, and the insulator **304** is or comprises intrinsic or lightly doped semiconductor material. The insulator **304** may, for example, be lightly doped relative to the cathode **302** and/or the anode **306**. The semiconductor material of the multilayer stacks may, for example, be or comprises polysilicon, monocrystalline silicon, germanium, indium gallium arsenide, or some other suitable semiconductor material. In some embodiments in which the multilayer stack is a MIM device, the cathode **302** and the anode **306** are or comprise metal or some other suitable conductive material and/or the insulator **304** is or comprises doped hafnium oxide, some other suitable metal oxide, or some other suitable insulator material.

In some embodiments, a thickness  $T_i$  of the insulator **304** is varied to adjust the threshold voltage of the unipolar selector **104**. For example, increasing a thickness of an insulator may increase a threshold voltage of the corresponding unipolar selector whereas decreasing the thickness may decrease the threshold voltage. In some embodiments, a doping concentration of the insulator **304** is varied to adjust the threshold voltage of the unipolar selector **104**. For example, increasing a doping concentration of an insulator

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may decrease a threshold voltage of the corresponding selector whereas decreasing the doping concentration may increase the threshold voltage. In some embodiments, a width  $W_1$  of the unipolar selector **104** is varied to adjust an “on” resistance of the unipolar selector **104**. For example, increasing a width of a selector may decrease an “on” resistance of the selector whereas decreasing the width may increase the “on” resistance.

With reference to FIG. 4, a schematic diagram **400** of some embodiments of the memory cell of FIG. 1 is provided in which a magnetic field generator **114** is coupled to the data-storage element **106**. The magnetic field generator **114** may be a current carrying wire configured to generate a magnetic field that can change polarity of the free ferromagnetic element **110** and thus change data state of the data-storage element **106**. In some embodiments, the magnetic field generator **114** is controlled by a controller and configured to generate an external magnetic field that resets the data-storage element **106** to the status opposite to the writing operation. An exemplary memory array application of the memory cell shown by FIG. 4 is given later with reference to FIG. 9A or FIG. 9B.

With reference to FIG. 5, a graph **500** of some embodiments of current-voltage (I-V) curves for the unipolar selector **104** of FIG. 1 is provided. A horizontal axis of the graph **500** corresponds to voltage, and a vertical axis of the graph **500** corresponds to current. Further, a right side of the graph **500** corresponds to the first polarity of the unipolar selector **104**, and a left side of the graph **500** corresponds to the second polarity of the unipolar selector **104**. The graph **500** includes a first I-V curve **502** where the bias  $V$  applied on the unipolar selector **104** is smaller than the threshold voltage  $V_t$ , and a second I-V curve **504** where the bias  $V$  applied on the unipolar selector **104** is greater than the threshold voltage  $V_t$ . As shown by the first I-V curve **502**, current is minimum when the voltage  $V$  applied on the unipolar selector **104** is smaller than the threshold voltage  $V_t$  of the unipolar selector **104**. As shown by the second I-V curve **504**, current increases when the voltage  $V$  applied on the unipolar selector **104** exceeds the threshold voltage  $V_t$  of the unipolar selector **104**.

With reference to FIG. 6A, a cross-sectional view **600a** of some embodiments of an integrated chip comprising the memory cell **102** of FIG. 1 is provided. The memory cell **102** overlies a substrate **602** and is located within an interconnect structure **604** that covers the substrate **602**. The interconnect structure **604** may be a back-end-of-line (BEOL) structure that comprises a plurality of wires **608** and a plurality of vias **610** surrounded by an interconnect dielectric layer **606**. The interconnect dielectric layer **606** may, for example, be or comprise silicon oxide, a low  $\kappa$  dielectric, some other suitable dielectric(s), or any combination of the foregoing. As used herein, a low  $\kappa$  dielectric may be, for example, a dielectric with a dielectric constant  $\kappa$  less than about 3.9. The wires **608** and the vias **610** are alternately stacked in the interconnect dielectric layer **606** to define conductive paths interconnecting components of the memory cell **102** and/or connecting the memory cell **102** to other devices (not shown) in the integrated chip. The wires **608** and the vias **610** may, for example, be or comprise metal, some other suitable conductive material(s), or any combination of the foregoing. For example, an intermediate via **610'** may define conductive paths electrically coupling the unipolar selector **104** in series with the data-storage element **106**. In some embodiments, due to the simplified structure of the disclosed selector, the memory cell **102** (including the unipolar selector **104** and the data-storage elements **106**) may be inserted

between two direct neighboring layers of metal wires. Therefore, the position of the memory cell **102** is more easily co-optimized with place and route requirement. The intermediate via **610'** may have a smaller height than the vias **610**.

With reference to FIG. **6B**, a cross-sectional view **600b** of some alternative embodiments of an integrated chip comprising the memory cell **102** of FIG. **1** is provided. Besides the similar features discussed above associated with FIG. **6A**, in some alternative embodiments, the memory cell **102** (including the unipolar selector **104** and the data-storage elements **106**) is inserted between two non-neighboring layers of metal wires **608**, and one or more additional wire layers may exist therebetween. The unipolar selector and the data-storage element **106** may be connected by multiple sub-vias **610b** connected by one or more isolated metal islands **608b**. The isolated metal islands **608b** has a width equal to that of the other wires **608** of the same interconnect layer and does not connect to other memory cells. The sub-vias **610b** may have a smaller greater height than the vias **610**. Here, two components are considered as "directly connected" if only conductive lines such as wires **608**, the isolated metal islands **608b**, and/or the vias **610** are used to connect the two components and no other electronic components are inserted therebetween. For example, the unipolar selector **104** and the data-storage element **106** are directly connected in this case.

With reference to FIG. **6C**, a cross-sectional view **600c** of some alternative embodiments of an integrated chip comprising the memory cell **102** of FIG. **1** is provided. Besides the similar features discussed above associated with FIGS. **6A-6B**, the unipolar selector **104** and the data-storage element **106** may also be directly stacked one above another. Thus, a bottom surface of the reference ferromagnetic element **108** of the data-storage element **106** and a top surface of the cathode **302** of the unipolar selector **104** may directly contact one another. Though not shown in figures, in some alternative embodiments, the free ferromagnetic element **110** of the data-storage element **106** may have a top/bottom surface directly contacting a top/bottom surface of the anode **306** or the cathode **302**, or the anode **306** may have a top/bottom surface directly contacting a top/bottom surface of the free ferromagnetic element **110** or the reference ferromagnetic element **108**. The cathode **302** and the anode **306** may have the same lateral dimensions. The free ferromagnetic element **110** and the reference ferromagnetic element **108** may have the same lateral dimensions. In some embodiments, the unipolar selector **104** has a lateral area between approximately 1 to 5 times a lateral area of the data-storage element **106**, which is a smaller size than an access transistor and at a higher quality than other types of bipolar selectors. Accordingly, the resulting memory cell is able to have relatively small size and good performance (e.g., high endurance and access speed). Lateral dimensions include length and width dimensions in parallel with surface of substrate. The memory cell **102** may be inserted between two neighboring or non-neighboring layers of metal wires **608**, and one or more additional wire layers may or may not exist therebetween.

With reference to FIG. **6D**, a cross-sectional view **600d** of some alternative embodiments of an integrated chip comprising the memory cell **102** of FIG. **1** is provided. Besides the similar features discussed above associated with FIGS. **6A-6C**, the isolated metal islands **608b** may be used as the cathode **302** of unipolar selector **104**. The unipolar selector **104** can also be reversed, and the isolated metal islands **608b** can be used as the anode **306**. In this case, the cathode **302**

and the anode **306** may have different lateral dimensions. The isolated metal islands **608b** has a width equal to that of the other wires **608** of the same interconnect layer. The insulator **304** therebetween may have the same width with one of the cathode **302** or the anode **306** not functioned by the isolated metal islands **608b** and greater than that of the isolated metal islands **608b**. By using the isolated metal islands **608b** as one electrode of the unipolar selector **104**, the manufacturing process is further simplified, and the device structure is more compact.

In some embodiments, the integrated chip is a standalone memory. In some alternative embodiments, the memory cell **102** is embedded in a logic circuit disposed on the substrate **602**. For example, a semiconductor device **612** is disposed on the substrate **602** integrating with the memory cell **102**. In some embodiments, the semiconductor device **612** is electrically coupled to the memory cell **102** by the wires **608** and the vias **610**. The semiconductor device **612** may, for example, be a metal-oxide-semiconductor (MOS) device, an insulated-gate field-effect transistor (IGFET), or some other suitable semiconductor device. In some embodiments, the semiconductor device **612** comprises a pair of source/drain regions **614**, a gate dielectric layer **616**, and a gate electrode **618**. The source/drain regions **614** are in the substrate **602**, along a top surface of the substrate **602**. The gate dielectric layer **616** and the gate electrode **618** are stacked over the substrate **602**, vertically between the substrate **602** and the interconnect structure **604** and laterally between the source/drain regions **614**.

With reference to FIG. **7**, a schematic view **700** of some embodiments of a memory array **702** comprising a plurality of memory cells **102** in a plurality of rows and a plurality of columns is provided. The memory cells **102** respectively comprises the unipolar selector **104** electrically coupled in series with the data-storage elements **106**. The memory cells **102** may, for example, each be as illustrated and described with regard to FIGS. **1-3**. As an example, bit lines (e.g.  $BL_m$ ,  $BL_{m+1}$ ,  $BL_{m+2}$ ) extend laterally along corresponding rows of the memory array and electrically couple with memory cells in the corresponding rows, whereas source lines (e.g.  $SL_n$ ,  $SL_{n+1}$ , and  $SL_{n+2}$ ) extend laterally along corresponding columns of the memory array and electrically couple with memory cells in the corresponding columns. The subscripts identify corresponding rows or columns and  $m$  or  $n$  is an integer variable representing a row or a column in the memory array **702**. Example numbers of  $m$  or  $n$  are 256, 512, 1024, etc. By appropriately biasing a bit line and a source line, the memory cell at the cross point of the bit line and the source line may be selected for reading or writing.

With reference to FIG. **8A-8B**, schematic diagram **800A-800B** of some embodiments of the memory array **702** of FIG. **7** are provided at various operational states to illustrate operation of the memory array **702**. The memory array **702** may be used as a read only memory device for which all memory cells are pre-set to a first data state (e.g., a logic "0") in a test stage or a specific environment. Once selected, a memory cell can only be written to a second data state (e.g., a logic "1") by a current flowing through the memory cell or a voltage applying across the memory cell with single polarity. The memory array **702** may be pre-set by an external magnetic field or an on-board magnetic field generator. FIG. **8A** illustrates the memory array **702** while writing a selected memory cell **102s** to a second data state (e.g., a logic "1"). FIG. **8B** illustrates the memory array **702** while reading a state of the selected memory cell **102s**.

As illustrated by FIG. **8A**, the selected memory cell **102s** is at the cross point of the source line  $SL_{n+2}$  and the bit line

$BL_{m+2}$ , for example. The bit line  $BL_{m+2}$  is biased with a writing voltage  $V_w$  while the source line  $SL_{n+2}$  is grounded. The writing voltage  $V_w$  is positive from the bit line  $BL_{m+2}$  to the source line  $SL_{n+2}$ , and provides a bias to the unipolar selector **104** exceeding the threshold voltage of the unipolar selector **104**, such that the selected memory cell **102s** is at a first polarity and the unipolar selector **104** of the selected memory cell **102s** is “on”. Current  $I_w$  flows through the selected memory cell **102s** and sets the data-storage element **106** of the selected memory cell **102s** to the second data state (e.g., a logic “1”). For example, the data-storage element **106** can be a MTJ structure and can be written by spin-transfer torque induced by the current  $I_w$ .

In some embodiments, other unselected memory cells **102u** are reversely biased by an inhibiting voltage at a second polarity opposite to the first polarity to keep the unselected memory cells “off” when writing the selected memory cell **102s**. The inhibiting voltage may have an absolute value that is equal to that of the writing voltage  $V_w$  or some other fractions of the writing voltage  $V_w$ . Alternatively, the inhibiting voltage may have an absolute value that is greater than the writing voltage  $V_w$ . For example, some unselected memory cells **102u** share the source line  $SL_{n+2}$  or the bit line  $BL_{m+2}$  with the selected memory cell **102s**. However, corresponding bit lines  $BL_m$ ,  $BL_{m+1}$  and source lines  $SL_n$ ,  $SL_{n+1}$  connecting these unselected memory cells **102u** are oppositely biased. For example, the source line  $SL_{n+2}$  and the bit line  $BL_m$  are both grounded, and the source line  $SL_n$  and the bit line  $BL_{m+2}$  are both biased at  $V_w$  or some other fraction of the writing voltage  $V_w$ . Accordingly, current flowing through the unselected memory cells **102u** is reduced or prevented, and writing disturbance to the unselected memory cells **102u** is reduced. In some embodiments, source lines  $SL_n$ ,  $SL_{n+1}$  are biased with the writing voltage  $V_w$  or some other fraction of the writing voltage  $V_w$  while bit lines  $BL_m$ ,  $BL_{m+1}$  can be grounded. Thus, leakage current and its resulting writing disturbance can be reduced.

As illustrated by FIG. 8B, the bit line  $BL_{m+2}$  is biased with a read voltage  $V_r$  while source line  $SL_{n+2}$  is grounded. The read voltage  $V_r$  is smaller than the writing voltage  $V_w$  and is small enough that the resulting read current  $I_r$  does not change a state of the selected memory cell **102s**. For example, the writing voltage  $V_w$  can be in a range of about 0.3V to about 1V, and the reading voltage  $V_r$  can be smaller than 0.3V such that the state of the selected memory cell **102s** is not altered. The reading voltage  $V_r$  also needs to be able to turn on the unipolar selector **104** of the selected memory cell **102s** such that current can flow through the data-storage element **106**. For example, the reading voltage  $V_r$  may need to be equal or greater than 0.1V. The read current  $I_r$  can be used to decide the resistances of the data-storage elements **106** and the corresponding data states of the selected memory cell **102s**. Similar to the writing operation, unselected memory cells **102u** are biased at a second polarity opposite to the first polarity when applying the reading voltage  $V_r$  across the selected memory cell **102s**. The unselected memory cells **102u** may be reversely biased by an inhibiting voltage having an absolute value that is equal to that of the reading voltage. Thus, leakage current and its resulting reading disturbance can be reduced.

With reference to FIG. 9A and FIG. 9B, schematic views **900a**, **900b** of the memory array **702** comprising a magnetic field generator **114** coupled to the plurality of memory cells **102** is provided according to some further embodiments. In addition to the description above related to FIG. 7 and FIGS. 8A-B, the magnetic field generator **114** may be arranged next to the plurality of memory cells **102** to provide an

external magnetic field to set or reset status of the plurality of memory cells **102**. As shown in FIG. 9A, the magnetic field generator **114** may be a current carrying wire arranged alongside the plurality of memory cells **102**. As shown in FIG. 9B, the magnetic field generator **114** may comprise a plurality of current carrying wires that can be separately controlled such that the memory cells **102** can be reset by groups, or even individually.

With reference to FIG. 10, a schematic view **1000** of some embodiments of a three dimensional (3D) memory array comprising a first memory array **702a** and a second memory array **702b** is provided. The first memory array **702a** and the second memory array **702b** are stacked, such that the second memory array **702b** overlies and is spaced from the first memory array **702a**. Stacking the first memory array **702a** and the second memory array **702b** may, for example, enhance memory density. In some embodiments, as illustrated, the first memory array **702a** and the second memory array **702b** are each as the memory array **702** in FIG. 7 is illustrated and described. As an example, anodes of the unipolar selectors **104** of the first memory array **702a** are connected to the first plurality of bit lines  $BL_m$ ,  $BL_{m+1}$ , and  $BL_{m+2}$ . Cathodes of the unipolar selectors **104** of the first memory array **702a** are connected to first terminals of the data-storage element **106** of the first memory array **702a**. Second terminals of the data-storage elements **106** of the first memory array **702a** are respectively connected to a first plurality of source lines  $SL_n$ ,  $SL_{n+1}$ , and  $SL_{n+2}$ . Anodes of the unipolar selectors **104** of the second memory array **702b** are connected to the second plurality of bit lines  $BL_m$ ,  $BL_{m+1}$ , and  $BL_{m+2}$ . Cathodes of the unipolar selectors **104** of the second memory array **702b** are connected to first terminals of the data-storage elements **106** of the second memory array **702b**. Second terminals of the data-storage elements **106** of the second memory array **702b** are respectively connected to a second plurality of source lines  $SL_n$ ,  $SL_{n+1}$ , and  $SL_{n+2}$ . Alternatively, it is appreciated by person in the art that the unipolar selectors **104** and the data-storage elements **106** of the first memory array **702a** and the second memory array **702b** can respectively arranged in mirror.

With reference to FIG. 11, a schematic view **1100** of some alternative embodiments the 3D memory array of FIG. 10 is provided in which the first memory array **702a** and the second memory array **702b** share source lines. As above, the source lines are respectively labeled  $SL_n$ ,  $SL_{n+1}$ , and  $SL_{n+2}$ , where the subscripts identify corresponding columns and  $n$  is an integer variable representing a column in the 3D memory array. Example numbers of  $m$  or  $n$  are 256, 512, 1024, etc. As an example, anodes of the unipolar selectors **104** of the first memory array **702a** are connected to the first plurality of bit lines  $BL_m$ ,  $BL_{m+1}$ , and  $BL_{m+2}$ . Cathodes of the unipolar selectors **104** of the first memory array **702a** are connected to first terminals of the data-storage element **106** of the first memory array **702a**. Anodes of the unipolar selectors **104** of the second memory array **702b** are connected to the second plurality of bit lines  $BL_m$ ,  $BL_{m+1}$ , and  $BL_{m+2}$ . Cathodes of the unipolar selectors **104** of the second memory array **702b** are connected to first terminals of the data-storage elements **106** of the second memory array **702b**. Second terminals of the data-storage elements **106** of the first memory array **702a** and the second memory array **702b** are respectively connected to a plurality of shared source lines  $SL_n$ ,  $SL_{n+1}$ , and  $SL_{n+2}$ . Alternatively, it is appreciated by person in the art that the unipolar selectors **104** and the data-storage elements **106** of the first memory array **702a** and the second memory array **702b** can respectively arranged in mirror. Also, the first memory array **702a**

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and the second memory array **702b** can share a plurality of bit lines and respectively connected to individual source lines.

With reference to FIG. 12, a cross-sectional view **1200** of some embodiments of an integrated chip comprising a pair of stacked memory cells **102** from the 3D memory array of FIG. 10 is provided. The stacked memory cells **102** are at the same row and the same column in the 3D memory array. Further, a lower one of the stacked memory cells **102** is in the first memory array **702a** of FIG. 10, whereas an upper one of the stacked memory cells **102** is in the second memory array **702b** of FIG. 10. The stacked memory cells **102** overlie a substrate **602** and are surrounded by an interconnect dielectric layer **606** of an interconnect structure **604**. Further, wires **608** and vias **610** in the interconnect dielectric layer **606** electrically interconnect components of the stacked memory cells **102**.

With reference to FIG. 13, a cross-sectional view **1300** of some alternative embodiments of the integrated chip of FIG. 12 is provided in which the stacked memory cells **102** are instead from the 3D memory array of FIG. 11. Accordingly, the stacked memory cells **102** share a source line SL defined by one of the wires **608**.

With reference to FIGS. 14-17, a series of cross-sectional views **1400-1700** of some embodiments of a method for forming an integrated chip comprising a memory array is provided in which memory cells of the memory array comprise unipolar selector.

As illustrated by the cross-sectional view **1400** of FIG. 14, an interconnect structure **604** is partially formed over a substrate **602**. The substrate **602** may, for example, be a bulk silicon substrate, a silicon-on-insulator (SOI) substrate, or some other suitable substrate. The interconnect structure **604** comprises a first interconnect dielectric layer **606a**, a first wire **608a** defining a bit line BL, and a first set of vias **610a**. The first interconnect dielectric layer **606a** accommodates the first wire **608a** and the first vias **610a** and may, for example, be or comprise silicon oxide, a low  $\kappa$  dielectric, some other suitable dielectric(s), or any combination of the foregoing. A low  $\kappa$  dielectric may be, for example, a dielectric with a dielectric constant  $\kappa$  less than about 3.9, 3, 2, or 1. The first wire **608a** and the first vias **610a** are stacked in the first interconnect dielectric layer **606a**, such that the first vias **610a** overlie the first wire **608a**.

In some embodiments, semiconductor devices (not shown) are on the substrate **602**, between the substrate **602** and the interconnect structure **604**. In some embodiments, additional wires (not shown) and/or additional vias (not shown) are alternately stacked in the first interconnect dielectric layer **606a**, between the substrate **602** and/or the first wire **608a**. The additional wires and/or the additional vias may, for example, define conductive paths leading from semiconductor devices (not shown) on the substrate **602**. In some embodiments, a process for partially forming the interconnect structure **604** comprises: 1) depositing a lower interconnect portion of the first interconnect dielectric layer **606a** on the substrate **602**; 2) forming the first wire **608a** inset into the lower interconnect portion; 3) forming an upper interconnect portion of the first interconnect dielectric layer **606a** on the first wire **608a** and the lower interconnect portion; and 4) forming the first vias **610a** inset into the upper interconnect portion. Other processes for partially forming the interconnect structure **604** are, however, amenable.

As illustrated by the cross-sectional view **1500** of FIG. 15, a unipolar selector **104** is formed overlying the bit line BL and electrically coupled to the bit line BL by one of the first

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vias **610a**. The unipolar selector **104** comprises a cathode **302**, an insulator **304**, and an anode **306**. The insulator **304** is between the cathode **302** and the anode **306**, and the cathode **302** overlies the anode **306**. The cathode **302**, the insulator **304**, and the anode **306** may, for example, define a PIN diode, a MIM device, or some other multilayer device. In some embodiments in which the cathode **302**, the insulator **304**, and the anode **306** define a PIN diode, the cathode **302** is or comprise N-type semiconductor material, the anode **306** is or comprises P-type semiconductor material, and the insulator **304** is or comprise intrinsic or lightly doped semiconductor material. The insulator **304** may, for example, be lightly doped relative to the cathode **302** and/or the anode **306**. The semiconductor material for the cathode **302**, the insulator **304**, and the anode **306** may, for example, be or comprises polysilicon, monocrystalline silicon, or some other suitable semiconductor material. In some alternative embodiments in which the cathode **302**, the insulator **304**, and the anode **306** define a MIM device, the cathode **302** and the anode **306** are or comprise metal or some other suitable conductive material and/or the insulator **304** is or comprises doped hafnium oxide, some other suitable metal oxide, or some other suitable insulator material.

In some embodiments, a process for forming the unipolar selector **104** comprises: 1) depositing an anode layer on the interconnect structure **604**; 2) depositing an insulator layer on the anode layer; 3) depositing a cathode layer on the insulator layer; and 4) patterning the multilayer film into the unipolar selector **104**. Other processes are, however, amenable. The depositing may, for example, be performed by chemical vapor deposition (CVD), physical vapor deposition (PVD), electroless plating, electroplating, some other suitable deposition process (es), or any combination of the foregoing. The patterning may, for example, be performed by a photolithography/etching process and/or some other suitable patterning process (es).

As illustrated by the cross-sectional view **1600** of FIG. 16, the interconnect structure **604** is extended around the unipolar selector **104**. The extended interconnect structure **604** further comprises a second interconnect dielectric layer **606b**, an isolated metal islands **608b**, and a set of sub-vias **610b**. In some embodiments, the isolated metal islands **608b** is formed together with other metal wires of the same interconnect layer and does not connect to other memory cells. The second interconnect dielectric layer **606b** accommodates the isolated metal islands **608b** and the sub-vias **610b**, and may, for example, be as the first interconnect dielectric layer **606a** is described. The isolated metal islands **608b** and the sub-vias **610b** are stacked in the second interconnect dielectric layer **606b**, such that the isolated metal islands **608b** is electrically coupled to the unipolar selector **104** by one of the sub-vias **610b** under the isolated metal islands **608b** and one of the sub-vias **610b** overlies the isolated metal islands **608b**.

In some embodiments, a process for extending the interconnect structure **604** comprises: 1) depositing a lower interconnect portion of the second interconnect dielectric layer **606b**; 2) forming the isolated metal islands **608b** and sub-vias **610b** underlying the isolated metal islands **608b** by a dual damascene process setting into the lower interconnect portion; 3) forming an upper interconnect portion of the second interconnect dielectric layer **606b** on the isolated metal islands **608b** and the lower interconnect portion; and 4) forming a sub-vias **610b** overlying the isolated metal islands **608b** and inset into the upper interconnect portion. Other processes for extending the interconnect structure **604** are, however, amenable.

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Still as illustrated by the cross-sectional view **1600** of FIG. **16**, a data-storage element **106** is formed overlying the interconnect structure **604**, on one of the sub-vias **610b**. The data-storage element **106** may, for example, be an MTJ, a MIM stack, or some other suitable structure for data storage. In some embodiments in which the data-storage element **106** is an MTJ comprising a reference ferromagnetic element **108**, a free ferromagnetic element **110**, and a barrier element **112**. The barrier element **112** is non-magnetic and is sandwiched between the reference ferromagnetic element **108** and free ferromagnetic element **110**. The reference ferromagnetic element **108** and free ferromagnetic element **110** are ferromagnetic, and the free ferromagnetic element **110** overlies the reference ferromagnetic element **108** and the barrier element **112**. Alternatively, locations of the reference ferromagnetic element **108** and free ferromagnetic element **110** are switched.

In some embodiments, a process for forming the data-storage element **106** comprises: 1) depositing a reference layer on the interconnect structure **604**; 2) depositing a barrier layer on the reference layer; 3) depositing a free layer on the barrier layer; and 4) patterning the reference, barrier, and free layers into the data-storage element **106**. Other processes are, however, amenable. For example, the free layer may be deposited at 1) and the reference layer may be deposited at 3). The depositing may, for example, be performed by CVD, PVD, electroless plating, electroplating, some other suitable deposition process (es), or any combination of the foregoing. The patterning may, for example, be performed by a photolithography/etching process and/or some other suitable patterning process (es).

As illustrated by the cross-sectional view **1700** of FIG. **17**, the interconnect structure **604** is completed around the data-storage element **106**. The completed interconnect structure **604** comprises a third interconnect dielectric layer **606c**, a third wire **608c** defining a source line SL, and a third via **610c**. The third interconnect dielectric layer **606c** accommodates the third wire **608c** and the third via **610c**. Further, the third interconnect dielectric layer **606c** may, for example, be as the first interconnect dielectric layer **606a** is described. In some embodiments, a process for completing the interconnect structure **604** comprises: 1) depositing the third interconnect dielectric layer **606c**; and 2) simultaneously forming the third wire **608c** and the third via **610c** inset into the third interconnect dielectric layer **606c**. Other processes for extending the interconnect structure **604** are, however, amenable.

The method illustrated by FIGS. **14-17** may, for example, be employed to form the memory cell in any one of FIGS. **1-4**, the integrated chip in any one of FIG. **6A-6D**, **12** or **13**, or the memory array in any one of FIGS. **7-11**. Further, while the cross-sectional views **1400-1700** shown in FIGS. **14-17** are described with reference to a method, it will be appreciated that the structures shown in FIGS. **14-17** are not limited to the method and may stand alone without the method.

With reference to FIG. **18**, a block diagram **1800** of some embodiments of the method of FIGS. **14-17** is provided.

At **1802**, an interconnect structure is partially formed on a substrate, where the partially formed interconnect structure comprises a bit line wire and a via on the bit line wire. See, for example, FIG. **14**.

At **1804**, a unipolar selector is formed on the via, where an anode of the first unipolar selector faces the bit line wire. See, for example, FIG. **15**.

At **1806**, the interconnect structure is extended around the unipolar selector. See, for example, FIG. **16**.

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At **1808**, a data-storage element is formed and electrically coupled to the unipolar selector. See, for example, FIG. **16**.

At **1810**, the interconnect structure is formed around the data-storage element, where the completed interconnect structure comprises a source line wire overlying and electrically coupled to the data-storage element. See, for example, FIG. **17**.

With reference to FIG. **19**, a block diagram **1900** of some embodiments of a method of operating a memory device is provided.

At act **1902**, a memory device including a plurality of memory cells arranged in rows and columns is formed. The memory device can be formed by method described above associated with FIGS. **14-18** or by other applicable fabrication methods. The plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series. The memory device may include memory cells shown by FIGS. **1-4**, memory arrays shown by FIGS. **7-11**, or integrated chips shown by FIG. **6A-6D**, **12** or **13**.

At act **1904**, in some embodiments, the plurality of memory cells is pre-set to a first data state (e.g. logic "0"). In some embodiments, the pre-set operation is performed after wafer fabrication but before testing and onetime programming (OTP). The pre-set operation may be performed by applying a high external magnetic field (for example 0.5~5 tesla) to set all the memory cells by aligning magnetic directions of ferromagnetic elements. The pre-set operation can be performed off-board or on-board by a magnetic field generator. The magnetic field generator is configured to apply the high external magnetic field magnetic field that couples to the data storage elements of the memory cells and sets the data storage elements to the first data state.

At act **1906**, a writing voltage is applied across a first selected memory cell at a first polarity to turn on the unipolar selector of the first selected memory cell. The data-storage element of the first selected memory cell is written to a second data state (e.g. logic "1"). The data-storage element may be written by the current flowing through the data-storage element or the voltage applied across the data-storage element. FIG. **8A** shows an example schematic diagram when performing a writing operation. By using the unipolar selector, remaining unselected memory cells can be biased at a second polarity opposite to the first polarity to keep the unselected memory cells off and thus minimize leakage current. By reducing leakage current, writing disturbance can be reduced.

At act **1908**, a reading voltage is applied across a second selected memory cell at the first polarity to turn on the unipolar selector of the second selected memory cell such that a data state of the data-storage element of the second selected memory cell is read. FIG. **8B** shows an example schematic diagram when performing a reading operation. By using the unipolar selector, remaining unselected memory cells can be biased at a second polarity opposite to the first polarity to keep the unselected memory cells off. Thus, leakage current and its resulting reading disturbance can be reduced.

At act **1910**, in some embodiments, the plurality of memory cells is reset to the first data state (e.g. logic "0") when needed. The reset operation may be performed by an on-board magnetic field generator, which can be a current carrying wire for example. The magnetic field generator is configured to generate an external magnetic field that couples to the data storage elements of the memory cells and sets the data storage elements to the first data state.



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While the block diagram **1800** of FIG. **18** and the block diagram **1900** of FIG. **19** are illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated orderings of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

In some embodiments, the present application provides a method of forming a memory device. The method includes forming a lower part of an interconnect structure over a substrate and forming a unipolar selector over the lower part of the interconnect structure. The method further includes forming a data-storage element over the unipolar selector and electrically coupled in series with the unipolar selector, the data-storage element having a variable resistance. The method further includes generating an external magnetic field by a magnetic field generator to pre-set the data-storage element to a first data state.

In some alternative embodiments, the present application provides a method of forming an integrated chip. The method includes forming a first memory array disposed over a substrate and comprising a first plurality of memory cells arranged in rows and columns. The first plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series with a current path across the data-storage element in one way. The first plurality of memory cells is absent of any selector coupled to the data-storage element with an opposite current path across the data-storage element in the other way. The method further includes forming a first plurality of bit lines extending along corresponding rows of first the memory array and respectively connected to first terminals of the first plurality of memory cells in the corresponding rows and forming a first plurality of source lines extending along corresponding columns of the first memory array and respectively connected to second terminals of the first plurality of memory cells in the corresponding columns. The method further includes forming a magnetic field generator coupled to the first memory array and configured to generate an external magnetic field to reset the data-storage element to a first data state.

In some alternative embodiments, the present application provides a method of operating a memory device. A memory array is provided comprising a plurality of memory cells arranged in rows and columns. The plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series. A writing voltage is applied across a first selected memory cell at a first polarity to turn on the unipolar selector of the first selected memory cell such that the data-storage element of the first selected memory cell is written to a first data state. A reading voltage is applied across a second selected memory cell at the first polarity to turn on the unipolar selector of the second selected memory cell. A data state of the data-storage element of the second selected memory cell is read without being altered.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes

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and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of forming an integrated chip, comprising: forming a first memory array over a substrate and comprising a first plurality of memory cells arranged in rows and columns, wherein the first plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series with a current path across the data-storage element in one way, wherein the data-storage element is configured to be set from a first data state to a second data state by applying a writing voltage across the data-storage element; forming a first plurality of bit lines extending along corresponding rows of first the memory array and respectively connected to first terminals of the first plurality of memory cells in the corresponding rows; forming a first plurality of source lines extending along corresponding columns of the first memory array and respectively connected to second terminals of the first plurality of memory cells in the corresponding columns; and forming a magnetic field generator coupled to the first memory array and configured to generate an external magnetic field to set the data-storage element to the first data state.
2. The method according to claim 1, wherein forming the magnetic field generator comprises forming a current carrying wire coupled to a controller and configured to generate a magnetic field that simultaneously resets the data-storage element of each of the first plurality of memory cells.
3. The method according to claim 1, wherein the data-storage element is a magnetic tunnel junction (MTJ) comprising a reference ferromagnetic element and a free ferromagnetic element separated by a tunneling barrier layer.
4. The method according to claim 1, further comprising: forming a second memory array stacked over the first memory array and comprising a second plurality of memory cells arranged in rows and columns, wherein the second plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series; forming a second plurality of bit lines extending along corresponding rows of the second memory array and electrically coupled with first terminals of the second plurality of memory cells in the corresponding rows; and forming a second plurality of source lines extending along corresponding columns of the second memory array and respectively connected with second terminals of the second plurality of memory cells in the corresponding columns.
5. The method according to claim 4, wherein the first memory array and the second memory array are embedded in a logic circuit disposed on the substrate.
6. The method according to claim 1, further comprising: forming a second memory array comprising a second plurality of memory cells arranged in rows and columns, wherein the second plurality of memory cells

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respectively comprises a unipolar selector and a data-storage element electrically coupled in series; and forming a second plurality of bit lines extending along corresponding rows of the second memory array and electrically coupled with first terminals of the second plurality of memory cells in the corresponding rows; and

wherein the first plurality of source lines also extends along corresponding columns of the second memory array and electrically coupled with second terminals of the memory cells of the second memory array in the corresponding columns.

7. The method according to claim 6,

wherein anodes of unipolar selectors of the first memory array are connected to the first plurality of bit lines, and wherein cathodes of the unipolar selectors of the first memory array are connected to first terminals of the data-storage element of the first memory array;

wherein the anodes of the unipolar selectors of the second memory array are connected to the second plurality of bit lines, and wherein the cathodes of the unipolar selectors of the second memory array are connected to first terminals of the data-storage elements of the second memory array; and

wherein second terminals of the data-storage elements of the first memory array and the second memory array are respectively connected to the first plurality of source lines.

8. A method of forming an integrated chip, comprising: forming a memory array over a substrate and comprising a plurality of memory cells arranged in rows and columns, wherein the plurality of memory cells respectively comprises:

a lower part of an interconnect structure over the substrate;

a unipolar selector over the lower part of the interconnect structure; and

a data-storage element over the unipolar selector and electrically coupled in series with the unipolar selector, the data-storage element having a variable resistance; and

generating an external magnetic field by a magnetic field generator to set the data-storage element to a first data state,

wherein the plurality of memory cells is configured to be written from the first data state to a second data state by a single polarity writing voltage applied across the data-storage element and the unipolar selector.

9. The method according to claim 8, further comprising: forming a bit line wire in the lower part of the interconnect structure; and

forming a via on the bit line wire,

wherein the unipolar selector is formed with an anode connected to the bit line wire through the via and a cathode connected to one end of the data-storage element.

10. The method according to claim 9, further comprising forming a source line wire overlying the data-storage element and connected to the other end of the data-storage element.

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11. The method according to claim 8, wherein the unipolar selector and the data-storage element are electrically coupled with a current path across the data-storage element in one way, wherein the plurality of memory cells is absent of any selector coupled to the data-storage element with an opposite current path across the data-storage element in the other way.

12. The method according to claim 11, wherein the data-storage element is configured to be read by applying a reading voltage across the data-storage element and the unipolar selector.

13. The method according to claim 8, wherein the data-storage element is a magnetic tunnel junction (MTJ) formed with a reference ferromagnetic element and a free ferromagnetic element separated by a tunneling barrier layer.

14. The method according to claim 13,

wherein a cathode of the unipolar selector is directly connected to the reference ferromagnetic element of the MTJ;

wherein an anode of the unipolar selector is directly connected to a source line; and

wherein the free ferromagnetic element of the MTJ is directly connected to a bit line.

15. The method according to claim 8, wherein the unipolar selector is a PIN diode.

16. A method of forming an integrated chip, comprising: forming a memory array comprising a plurality of memory cells arranged in rows and columns, wherein the plurality of memory cells respectively comprises a unipolar selector and a data-storage element electrically coupled in series; and

generating an external magnetic field by a magnetic field generator to set the data-storage element to a first data state,

wherein the unipolar selector and the data-storage element are electrically coupled with a current path across the data-storage element in one way, wherein the plurality of memory cells is absent of any selector coupled to the data-storage element with a opposite current path across the data-storage element in the other way, and wherein the plurality of memory cells is configured to be written from the first data state to a second data state by applying a writing voltage of a first polarity across the data-storage element and the unipolar selector of a selected memory cell.

17. The method according to claim 16,

wherein unselected memory cells are biased at a second polarity opposite to the first polarity when applying the writing voltage across the selected memory cell.

18. The method according to claim 17, wherein the unselected memory cells are reversely biased by an inhibiting voltage having an absolute value that is equal to that of the writing voltage.

19. The method according to claim 16, wherein the magnetic field generator is off-board of the integrated chip.

20. The method according to claim 16, wherein the magnetic field generator is a current carrying wire that is magnetically coupled to the data-storage elements of the plurality of memory cells.

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